Dovan Darazi

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RESEARCH INTERESTS

High Energy Physics: Theory and Computational/Experimental, Astroparticle Physics, Early Universe Physics, Dark Matter Physics, Cosmology, General Relativity.

ACADEMIC HISTORY

American University of Beirut Jan 2025 – Present

Masters of Science in Physics

Electives: Quantum Field Theory, Advanced Mathematical Methods

Lebanese University, Faculty of Sciences II Oct 2021 – Jul 2024

Bachelor of Science in Physics

GPA: 3.6 (passing grade 60%)
Rank: first year: 1/17; second year: 1/3; third year: 1/1

Elective courses: Waves and vibrations, Fluid mechanics, Classical Field Theory, Laser Physics, Thermal Transfer, Material Science, Molecular Physics, Medical Imaging

RESEARCH EXPERIENCE

Master's Thesis: Investigating Baryogenesis through B-Meson Decays May 2026 – Present

American University of Beirut — BaBar data analysis with Dr. Racha Cheaib

- Investigating a possible mechanism for baryogenesis through B-meson decays involving a Ξ_c^+ baryon and a dark matter particle.
- Developing an analysis using BaBar data to search for this decay signature and explore its potential connection to the matter–antimatter asymmetry of the Universe.

Research assistant, American University of Beirut (AUB) Jun 2025 – Oct 2025

Laboratory for Plasma and Fluid Dynamics

- Designed, built, and tested an improved divertor model for magnetic confinement nuclear fusion devices aimed at filtering high-energy debris within the plasma. The design replaces conventional solid components with a flowing liquid metal system capable of self-healing after repeated particle impacts, significantly reducing material degradation and maintenance requirements. The system captures and removes debris from the plasma, filters it through the liquid medium, and recirculates the purified material back into the divertor, enhancing durability and operational efficiency.

Research assistant, American University of Beirut (AUB) Sep 2023 – Oct 2025

Laboratory for Plasma and Fluid Dynamics

- Project lead on developing a high-speed stereoscopic diagnostics device in order to detect and identify UFOs, Flakes and Dust particles in the plasma flow of a magnetic confinement nuclear fusion reactor. The objective of this project is to determine particles' sizes as well as their positions in three dimensions and their velocities in component form. This data plays a pivotal role in informing the necessary adjustments to the input parameters of the device, enabling us to anticipate and mitigate potential disruptions to the plasma flow with greater precision.
- **Access The Project Documents: here**

Intern, Lebanese American University (LAU) Jul 2023 – Sep 2023

- Assisted in data collection, organization, and analysis for a Research project on the water-energy-food nexus studying the impact of the Lebanese economic and energy crisis on water security. Worked on finalizing the paper for publication.

PUBLICATIONS AND PRESENTATIONS

Publications: Google Scholar

- Navigating the water–energy nexus amidst the Lebanese economic crisis

Presentations

- Presented my project on imaging disruptions in magnetic confinement fusion devices at the first MENA region plasma fusion physics (MPFu) conference in Gammarth, Tunisia, as the youngest presenter.

ACADEMIC AWARDS

Graduate Fellowship and Assistantship Program Jan 2025 – Present

American University of Beirut — Beirut, Lebanon

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant, American University of Beirut (AUB) Jan 2025 – Present

Teaching Physics labs for engineers and premeds (PHYS 211L & PHYS 206L)

- My duties as a TA were: Deliver a lecture on how to do the experiment, guide and assist the students while they're doing the experiment, correct reports that the students submit for each experiment, Design and correct quizzes that test the students' preparation, have office hours for followups, questions, and inquiry, correct final exams along with other teaching assistants.

Particle Physics Summer School Instructor and Mentor Jul 2026 – Aug 2026

Summer Research Experience Program (SREP): Analyzing LHCb Data

- Mentored a group of high school students, teaching the fundamentals of particle physics, detector physics, Python programming, and large-scale data analysis.
- Guided students in analyzing LHCb B-meson decay data, including particle identification and developing code to reconstruct specified decay channels using charge and four-momentum conservation.
- Supervised the selection and optimization of analysis cuts by evaluating the trade-off between efficiency and purity, culminating in a measurement of the asymmetry between B^+ and B^- decays.
- Supported students in presenting their methods and findings to fellow participants and physics faculty at the conclusion of the program.

Particle Physics Summer School Instructor and Mentor Jul 2026 – Aug 2026

Summer Research Experience Program (SREP): Designing a Dark Photon Detector

- Mentored a group of high school students, teaching the fundamentals of particle physics, detector physics, Python programming, and large-scale data analysis.
- Guided students in exploring different types of particle detectors and their operating principles, applying this knowledge to develop a conceptual detector design for a research objective of their choice.
- Supervised the team's development of a conceptual detector designed to search for dark photons.

TECHNICAL SKILLS

Python	Advanced	**LaTeX**	Intermediate
ROOT	Intermediate	**OriginLab**	Intermediate
C++	Intermediate	**AutoCAD**	Intermediate
Rust	Intermediate	**Inventor**	Intermediate
MATLAB	Intermediate	**Blender**	Beginner
Linux	Intermediate	**COMSOL Multiphysics**	Beginner
SQL	Beginner	**Raspberry Pi**	Beginner

PROJECTS

Monte Carlo Simulation and Analysis of the Ising Model Apr 2025 – Jun 2025

Statistical Mechanics — Final Course Project

- Studied the Ising model and implemented a Monte Carlo simulation to investigate its statistical mechanical behavior.
- Analyzed the simulation results alongside the theoretical description of the model as part of the course's final project.

Investigating High-Efficiency Thermoelectric Metals for Integration with Hollow Blocks Dec 2024 – Jan 2025

Numerical Analysis II course, Lebanese University, Faculty of Sciences II, Department of Physics

- In this project, we aim to design an innovative system for sustainable energy generation by leveraging the temperature gradient between the interior and exterior of buildings. Working as a team of two, we are researching thermoelectric metals with high Seebeck coefficients for integration into hollow blocks (bricks). Our goal is to identify materials that are not only efficient in converting heat to electricity but are also cost-effective and abundant, ensuring scalability and accessibility for widespread use.

  To achieve this, we are utilizing COMSOL Multiphysics to create detailed simulations of the proposed device, enabling us to model and optimize its performance under real-world conditions. By combining advanced material analysis with cutting-edge simulation tools, this project has the potential to pave the way for practical applications of thermoelectric technology in building materials, contributing to energy-efficient infrastructure and renewable energy solutions.

GOLEM Tokamak: Toroidal Field Experiment Summer 2024

Virtually operated GOLEM tokamak, Czech Technical University in Prague

- Investigated the impact of toroidal magnetic field strength on plasma stability and confinement by varying the toroidal field capacitor voltage.
- Studied its effects on key plasma parameters, including plasma current, edge safety factor, discharge duration, and electron energy confinement time.

Developing an Interactive Virtual Assistant Display Mar 2024 – Jun 2024

Open Laboratory, Lebanese University, Faculty of Sciences II, Department of Physics

- Developing a 3D interface for the laboratory's virtual assistant using Blender, with the goal of enhancing user interaction and visualization. The interface is designed to be displayed on an anamorphic screen, providing a dynamic and immersive visual experience. This project serves as a foundation for creating a more advanced projection system, aimed at improving display technologies for laboratory applications and beyond.

Studying the Barber pole effect in sucrose solution Oct 2023 – Feb 2024

Optics Laboratory Project, Lebanese University, Faculty of Sciences II

- As part of a student project supervised by a professor at the Laboratory of Biomedical Imaging (LBMI), Lebanese University, Faculty of Sciences II, our team conducted a series of experiments to investigate the barber pole effect. The project involved designing and performing experiments to observe the phenomenon and collect relevant experimental data.

  Through comprehensive data analysis, we validated the theoretical concepts identified during the literature review, demonstrating alignment between our findings and existing research. The results were presented to the supervising professor and peers, showcasing our approach, methodology, and key insights. This project provided valuable hands-on experience in experimental design, data analysis, and effective scientific communication.

Studying and Using a Prism Spectrometer Goniometer Oct 2023 – Jan 2024

Optics Laboratory Course, Lebanese University, Faculty of Sciences II

- Calibrated a non-digital spectrometer goniometer using a mercury lamp to ensure accuracy in subsequent measurements. The calibrated setup was utilized to determine the Balmer spectra of a hydrogen lamp and calculate the Rydberg constant.

  A theoretical geometrical optical analysis was conducted to achieve the best fit for the measured data, with results visualized using Python's matplotlib library. To further enhance understanding, the studied concept was modeled in Blender, providing a 3D representation of the optical system. This project integrated experimental techniques, computational analysis, and 3D modeling to deepen insights into spectroscopic phenomena.

The Effects of Photon Energy and Magnetic Fields on the Germination/Growth of Plants Aug 2023 – Oct 2023

Student-led project under the supervision of a professor (LBMI, Lebanese University, Faculty of Sciences II)

- Designed and optimized experimental methods for seed germination and growth under controlled conditions. The experiments investigated the effects of blue and red light (individually) on germination, as well as the influence of a magnetic field on the process.

  The study aimed to explore the potential role of quantum processes in seed germination and growth, providing insights into how environmental factors such as light wavelength and magnetic fields impact these biological mechanisms. This project combined experimental precision with theoretical exploration to advance understanding in biophysics and plant sciences.

WORKSHOPS, CONFERENCES, AND SCHOOLS

Physics Without Frontiers: Lebanon's First Quantum Technology Workshop Oct 2025

Abdus Salam International Centre for Theoretical Physics

STARS: An Introduction to their Structure and Evolution Sep 2025

The fourth MPFuS Summer 2024

Attended the fourth MPFuS summer school for plasma and fusion physics held in Gammarth, Tunisia.

STUDIED TOPICS

Physics Particle Physics, Quantum Mechanics, Quantum Field Theory, Nuclear Physics, Classical Field Theory, General Relativity, Computational Physics, Plasma Physics, Fluid Mechanics, Experimental Physics, Electric Circuits, Error Analysis, Classical Mechanics, Statistical Mechanics, Electromagnetism, Waves And Vibrations, Laser Physics, Thermal Transfer, Material Science, Molecular Physics, Medical Imaging Physics.

Mathematics Tensor Analysis, Linear Algebra, Differential Geometry, Topology, Group Theory, Complex Analysis, Fourier Series and Fourier Transform, Laplace Transform, Ordinary and Partial Differential Equations, Numerical Computing and Analysis, Real Analysis

Space Science Astronomy, Astrophysics, Planetary Geology

ONLINE COURSES

coursera.org

Particle Physics: An Introduction, From the Big Bang to Dark Energy, Astro 101: Black Holes, Astronomy: Exploring Time and Space, Astrobiology: Exploring Other Worlds, The Science of the Solar System, How to Write and Publish a Scientific Paper

CERTIFICATIONS

- Raspberry Pi Certification from IEEE
- First Aid Certification from Red Cross

LANGUAGES

English Fluent

Arabic Native

French Conversational